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Falcon 9 CRS-10 Vehicle Simulator and Navigation

Topics in Aerospace Engineering - Team 5

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1 Introduction and Project Scope

1.1 Project 2 objectives

Project 2 develops a physics-based Falcon 9 simulator and a navigation system driven by simulated sensor measurements. Forces, moments, changing mass properties, and numerical integration generate the vehicle motion; the navigation algorithm estimates states from imperfect observations of that motion. The course identifies this environment as the computational foundation for guidance design in Project 3 and integrated control in Project 4 [11, slides 2–4].

The assignment covers equations of motion and coordinate systems, gravity and atmosphere, propulsion and aerodynamics, mass/inertia variation, configuration changes, and estimation of position, velocity, attitude, and angular velocity. Its validation requirement combines representative flight phases, public flight-data comparisons, and navigation-error analysis [11, slides 4–6]. Accordingly, the report distinguishes physical-model verification, agreement with external flight evidence, and estimation error relative to simulation truth. These comparisons establish different aspects of the simulator's credibility.

1.2 CRS-10 selection and vehicle configuration

CRS-10 is SpaceX's tenth contracted Dragon cargo-resupply mission to the International Space Station. The official mission material identifies a Falcon 9/Dragon launch from Launch Complex 39A, first-stage recovery at Landing Zone 1, and an approximate sequence of cutoff, separation, and restart events [1]. The public technical webcast and its published ascent digitization provide an accessible reference for a quantitative comparison [13, 14].

The mission exercises three connected trajectory segments: common-stack ascent, first-stage return, and upper-stage continuation. It therefore tests configuration changes and inherited states as well as continuous motion. The Dragon payload also fixes the configuration choice: the model carries one loaded Dragon/trunk assembly rather than a generic satellite and fairing. NASA's 2490 kg cargo inventory is a subset of that loaded assembly, not its total mass [2].

Figure 1 identifies the selected hardware and official mission document. The press kit supplies mission context and approximate preflight timing; generic Falcon user guides supply supplementary geometry and operating-point evidence with their configuration/date limitations. Internal dimensions and mass properties require the evidence-qualified modeling described in the vehicle chapter.



Figure 1 CRS-10 reference configuration. Left: Falcon 9 and Dragon at LC-39A before launch, NASA/Glenn Benson [12]. Right: header and mission-patch excerpts from the official SpaceX press kit [1]. The images identify the hardware and source; they do not determine the internal mass distribution.

1.3 Simulation and validation scope

The initial vehicle contains both stages and the loaded Dragon assembly. At separation, a conservative state map creates two children. The first stage proceeds through boostback, coast, entry burn, aerodynamic descent, landing burn, and first geometric contact. The upper child retains Dragon and continues under its own engine and attitude commands. Both children inherit the propagated separation state; a separately selected descent checkpoint does not create the full-return trajectory.

Flight commands provide a repeatable scenario for exercising the physical models. Scheduled reference times are distinguished from state-triggered return conditions, and project-designed return commands are classified as assumptions. The upper continuation demonstrates independent propagation rather than validated orbital insertion. First-stage navigation propagates its own quaternion from noisy gyros in a multiplicative extended Kalman filter (MEKF) with GNSS/LiDAR updates. A simulated star tracker supplies gated attitude observations; acceptance is checked per run. Physical attitude supplies sensor generation, an assumed initial alignment, and the evaluation reference; it is not continuously supplied to the active estimator.

External trajectory comparison uses the retained video-derived ascent observations over approximately 0–140 s. Their classification is E: they are digitized public readouts, not released onboard telemetry. The ascent operating point and pitch timescale were adjusted using the same record, so their agreement is calibration evidence. Model-level tests, continuous state transfer, return/upper trajectories, and navigation errors provide additional evidence with their own stated assumptions. Navigation error against the simulator's truth does not by itself establish the accuracy of the physical vehicle model.

2 Simulation Architecture and Shared Conventions

2.1 Physical truth, measurements, and estimates

The physical branch computes loads and mass properties and integrates vehicle motion. The measurement branch samples that truth through sensor error and availability models. Navigation propagates its own attitude and estimates the IMU-station motion. Comparison uses the same station and time. Figure 2 separates these interfaces.

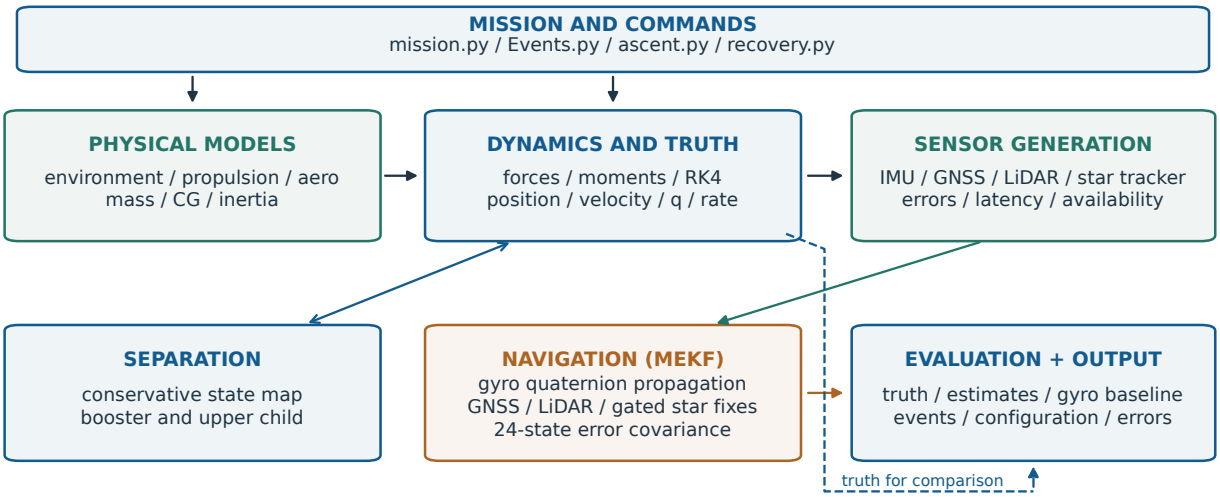


Figure 2 Implemented data flow. The MEKF uses simulated sensor measurements and its own propagated attitude. Truth supplies sensor generation and evaluation; initial alignment is assumed. Return commands use physical-state feedback, while navigation is evaluated separately.

The mission layer selects the initial configuration, commands, and accepted phase. A throttle command becomes thrust and flow through propulsion; an aerodynamic configuration and fin command become force and moment through aerodynamics. The dynamics model then calculates the resulting motion. Neither the force provider nor the mass model prescribes the reference altitude/speed history. Table 1 assigns the corresponding implementation responsibilities.

Function	Main modules	Interface

Vehicle properties	`VehicleData.py`
	`vehicle_mass_model.py`
	`mass_properties.py`
		Trial inventory to mass, CG, inertia, and derivatives.

Environment and loads	`Environement.py`
	`propulsion.py`
	`Aerodynamics.py`
	`load_interface.py`
		Trial state/commands to loads and exhaust data.

Motion and configuration	`dynamics.py`
	`frames.py`
	`staging.py`
		Derivative, integrated state, and inherited child states.

Measurement and estimation	`sensors.py` `kalman.py`	Sensor-station truth to observations and separate estimates.

Mission execution	`main.py`
	`mission.py`
	`Events.py`
		Commands, accepted clocks/events, state synchronization, and saved output.

Table 1 Responsibilities of the shared implementation. Configuration and scenario decisions are kept separate from reusable component arithmetic and equations of motion.

2.2 Frames, origins, and attitude convention

Frame B is right-handed and fixed to the vehicle, with $+x_B$ toward the nose. Frame E is Earth-centred, launch-aligned, and rotating with Earth. Code identifiers call it ECEF, but its axes are not conventional

Greenwich-axis ECEF. At the launch site, $+x_E$ is north, $-y_E$ east, and $+z_E$ up; the initial structural base is $[0, 0, R_E]^\mathsf{T}$ with $R_E = 6371000\,\mathrm{m}$.

The scalar-first Hamilton quaternion $\boldsymbol{q}_{BE} = [q_w, q_x, q_y, q_z]^\mathsf{T}$ maps Body components into E using C_{BE}. Equation (1) defines the vector transformation and launch orientation:

$$\boldsymbol{u}^E = \mathrm{C}_{BE}\boldsymbol{u}^B, \qquad \boldsymbol{q}_{BE}(0) = \begin{bmatrix}\sqrt{1/2} & 0 & -\sqrt{1/2} & 0\end{bmatrix}^\mathsf{T}, \qquad \mathrm{C}_{BE}(0) = \begin{bmatrix}0 & 0 & -1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}. \tag{1}$$

Thus Body $+x$ is initially upward, Body $+y$ maps to $+y_E$, and Body $+z$ maps to $-x_E$. Quaternions are normalized, and $\boldsymbol{q}$ and $-\boldsymbol{q}$ denote the same orientation. The adopted latitude $\lambda = 28.6^\circ$ rotates the Earth-rate vector into the launch-aligned frame as $\boldsymbol{\Omega}^E = \Omega_E[\cos\lambda, 0, \sin\lambda]^\mathsf{T}$, with $\Omega_E = 7.292115 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$. It does not relocate the pad into Greenwich-axis coordinates.

Reference point and component basis are separate choices. Table 2 distinguishes them. A structural station $\boldsymbol{s}^B$ is measured from the stage base, whereas the corresponding CG force arm is $\boldsymbol{d}^B = \boldsymbol{s}^B - \boldsymbol{c}^B$, where $\boldsymbol{c}^B$ is the current CG station.

Quantity	Meaning
Dynamic position $\boldsymbol{r}_C^B$	Earth centre to instantaneous CG, expressed in current Body axes.
Dynamic velocity $\boldsymbol{v}_C^B$	CG velocity relative to rotating Earth, in current Body components.
Vehicle-object position/velocity	Structural-base state, with the same external origin and Body component basis.
Component/sensor station $\boldsymbol{s}^B$	Fixed structural location measured from the relevant stage base.
Moment and inertia	About the instantaneous vehicle CG, in Body axes.
Filter position/velocity	At the fixed IMU station, expressed in frame E.

Table 2 Shared origins and representations. Storing an Earth-centred vector in Body components does not make its origin the CG.

2.3 Physical and estimated states

The physical state contains 15 stored numbers: three CG-position components, three CG-velocity components, four quaternion components, three angular-rate components, and two stage-indexed propellant inventories. The unit-quaternion constraint leaves three independent attitude coordinates. Each separated child retains the same state layout, with the detached stage's inventory set to zero.

The active MEKF maintains nominal IMU-station position and velocity in E, Body-frame accelerometer bias, the unit quaternion, gyro bias, three gyro scale factors, and six off-diagonal gyro calibration terms. Its covariance represents a 24-component error state: position, velocity, accelerometer bias, small-angle attitude error, gyro bias, scale factors, and misalignment terms. The quaternion occupies four nominal numbers but only three attitude-error coordinates. The original nine-state, supplied-attitude mode remains an optional diagnostic; it is not the default run.

Noisy gyro samples propagate the estimated quaternion using the same Body-to-E convention as Equation (1). GNSS and LiDAR update the coupled estimate; the star-tracker correction is conditional on availability and an innovation gate. Measured rate, estimated gyro bias, and the calibration matrix form the angular-rate estimate; rate is not a separate dynamic filter state. An independent, unaided gyro quaternion provides a drift baseline. The default initial attitude equals the launch alignment, an A-class initialization assumption. The filter does not receive physical attitude during propagation or correction. Calibration-state estimation does not by itself demonstrate identifiability; the nominal run injects gyro bias and noise but zero scale-factor and misalignment errors.

The assumed first-stage stations are $[38, 0, 0]^\mathsf{T}\,\mathrm{m}$ for the IMU, $[41, 0, 0]^\mathsf{T}\,\mathrm{m}$ for GNSS, and $[0.5, 0, 0]^\mathsf{T}\,\mathrm{m}$ for LiDAR. These are A-class experiment settings. They remain on the structure when the CG moves or the upper stage separates. The ideal sensor state follows the structural-point kinematics; sensor errors are added afterward. An estimate at the IMU station is therefore compared with truth at that station, not with the changing geometric CG.

The stored angular rate and simulated ideal gyro use the Body-relative-to-E rate. A physical inertial gyro instead senses Body-relative-to-inertial rotation; the present sensor convention omits that Earth-rate difference. The star-tracker interface explicitly rotates between E and an inertial frame that coincides with E at launch. This conversion does not change the gyro convention. Translation includes Coriolis and centrifugal acceleration; a fully physical inertial strapdown model requires a corresponding gyro Earth-rate treatment.

The star tracker is an A-class simulated aiding sensor, not evidence of CRS-10 flight hardware. Its defaults are 4 Hz, 0.1 s latency, 5 arcsec cross-boresight and 40 arcsec boresight-roll standard deviations, a 60 km minimum altitude, and a 2 deg s^{-1} rate limit. The reviewed seed-1 full-flight run rejects all available star-tracker fixes at the innovation gate. Its measured performance therefore characterizes gyro/GNSS/LiDAR estimation. Sensor and filter chapters must distinguish an implemented correction interface from accepted aiding and assess covariance consistency alongside point-estimate errors.

2.4 Load and execution interfaces

The load record separates nongravitational force in newtons, CG moment in newton-metres, central gravity in m s^{-2}, rotating-frame acceleration in m s^{-2}, nonnegative stage consumption in kg s^{-1}, and nozzle-exit flow data. Station forces contribute $\boldsymbol{d}\times\boldsymbol{F}$ once; free aerodynamic/control moments are supplied separately. Gravity and frame acceleration are added once in the dynamics, rather than included again in the force sum.

At each RK4 trial evaluation, properties and loads are recomputed from the trial state without advancing accepted vehicle inventories, event order, or sensor random generators. The mission loop synchronizes accepted states and samples due sensors. Steps are shortened at relevant command, event, sensor, and output boundaries. At separation, the same first-stage sensor/filter objects continue while two physical children inherit the parent state. The upper child is propagated independently on the common mission clock. Detailed command laws and scheduling belong to mission execution; these interfaces define what the models exchange.

3 Vehicle Configuration, Dynamics, and Mission Definition

3.1 Configuration and evidence-qualified parameters

The attached vehicle comprises stage 1, stage 2, and a loaded Dragon/trunk assembly. After separation, the lower vehicle contains stage 1 and the upper vehicle contains stage 2 and Dragon. The parameter register distinguishes K (authoritative published information), E (literature/data-derived estimate), and A (explicit modeling assumption). E/A denotes a sourced estimate whose use also requires an assumption. Source authority and applicability to the selected 2017 flight are separate considerations.

Table 3 consolidates the configuration used by `vehicle_parameters.py` and `VehicleData.py`. The 62.4 m equivalent height is the sum of adopted stage/Dragon lengths; it is not presented as a measured CRS-10 overall dimension. Stage boundaries and the contiguous attachment datums remain conditional. The 8430 kg loaded Dragon mass is retained as a provisional external estimate; the NASA cargo inventory is included within it. No separate satellite fairing is added.

Parameter	Adopted value	Class	Basis
Outer diameter	3.66 m	K	Later official external geometry [5].
Stage-1 / stage-2 length	42.6/12.6 m	E/A	Legacy FT geometry and assumed station allocation [17].
Stage-1 dry / propellant mass	22200/410900 kg	E	Adopted research mass budget [7].
Stage-2 dry / propellant mass	4000/107500 kg	E	Same literature set [7].
Loaded Dragon assembly	8430 kg	E	Provisional external estimate [3].
Dragon equivalent length / diameter	7.2/3.66 m	E/A	Earlier Dragon geometry transferred to a uniform assembly [4].
Dragon / stage-2 CG station	58.8/48.9 m	A	Midpoints above assumed attached-stage datums.
Stage-1 nominal dry CG	17.521 m	E/A	Component budget and residual allocation.
Initial attached mass	553030 kg	E/A	Sum of the adopted inventories and structures.
Equivalent model height	62.4 m	E/A	Sum of the adopted axial envelopes.

Table 3 Consolidated vehicle inputs. Stations are measured from the attached stack base along Body $+x$. A known external diameter does not establish a known internal tank or mass distribution.

3.2 Propulsion and propellant depletion

Stage 1 has one central engine and eight surrounding engines on an assumed 1.3 m radius ring. Stage 2 has one MVac engine. The force-application pivots are 1.0 m and 0.5 m above the respective bases (A); pivot stations, engine mass-centre stations, and nozzle-exit stations are distinct. Engine dry masses are already included in the stage dry budgets.

For a running steady engine with throttle u, the implementation evaluates Equation (2); an off engine has zero thrust and flow:

$$T = uT_{\mathrm{vac}} - A_e p_{\mathrm{amb}}, \qquad Q = \frac{uT_{\mathrm{vac}}}{I_{\mathrm{sp,vac}} g_0}, \qquad g_0 = 9.80665\,\mathrm{m\,s^{-2}}. \tag{2}$$

Here T is thrust, Q positive propellant outflow, A_e nozzle exit area, p_{amb} ambient pressure, and $I_{\mathrm{sp,vac}}$ the adopted vacuum specific impulse. The code retains a fixed-area ambient-pressure correction; the steady throttle does not scale that pressure term. Smooth ignition/shutdown envelopes subsequently scale the evaluated thrust and flow together. The effective stage-1 vacuum point is $756000 + 0.68(101325) = 824901\,\mathrm{N}$ at full throttle, using the historical sea-level reference and a literature exit area. This mixed-source operating set is E/A, not a certified CRS-10 engine rating [6, 7]. Table 4 lists the adopted engine inputs.

Parameter	Stage-1 engine	Stage-2 engine
Engine count	9	1
Adopted vacuum thrust	824.901 kN (E/A)	935 kN (E)
Vacuum specific impulse	311 s (E)	348 s (E)
Exit area	0.68 m^2 (E)	4.90 m^2 (E)
Steady minimum throttle	0.60 (E)	0.40 (E)

Table 4 Engine operating inputs [7, 6]. The numerical scenario chooses engine count and throttle separately from these engine relations. A stored 5° gimbal parameter (A) does not establish implemented physical gimbal dynamics.

Stage-1 loading is $m_{f,0} = 123500$ kg of RP-1 and $m_{o,0} = 287400$ kg of LOX [9]. Their loaded ratio is 2.32713, while consumption uses $\eta = Q_o/Q_f = 2.35$ [8]. Equation (3) reconstructs both tanks from the stage's total consumed mass Δm:

$$Q_f = \frac{Q}{1+\eta}, \quad Q_o = \frac{\eta Q}{1+\eta}, \qquad m_f = m_{f,0} - \frac{\Delta m}{1+\eta}, \quad m_o = m_{o,0} - \frac{\eta \Delta m}{1+\eta}. \tag{3}$$

The adopted combination depletes LOX first and leaves 1202.128 kg of RP-1 at its mathematical limit. That residual is unusable under the assumed fixed ratio and is separate from a mission reserve. Reconstruction assumes fixed initial loading and no venting, refilling, or variable mixture schedule. Stage 2 retains an equivalent uniformly distributed propellant cylinder rather than the first-stage dual-column model.

3.3 Changing mass, CG, and inertia

3.3.1 Liquid and dry-component models

Each first-stage liquid is a settled, flat-ended cylindrical column at constant density. Table 5 lists its geometry and reference conditions. Structural section lengths are transferred to equivalent tank envelopes; domes, internal hardware displacement, and datum alignment are assumptions.

Quantity	RP-1	LOX
Inner radius [m], E/A	1.8269	1.8271
Bottom station [m], E/A	4.0	18.7
Equivalent section length [m], E/A	14.7	22.1
Density [kg m^{-3}], E	820	1250
Reference temperature [K], E	266	66

Table 5 Equivalent tank inputs. Radii subtract the modeled skin thickness from 1.83 m; section lengths/stations use Ecker's structural model [15, Table 1]. Density/temperature pairs use Miquel's Table 5, whose reference pressure is unspecified [16].

For liquid mass m_i, density ρ_i, radius r_i, bottom station b_i, and outflow Q_i, Equation (4) defines height, centroid, and centroid rate:

$$h_i = \frac{m_i}{\rho_i \pi r_i^2}, \qquad c_i = b_i + \frac{h_i}{2}, \qquad \dot{c}_i = -\frac{Q_i}{2\rho_i \pi r_i^2}. \tag{4}$$

The height must remain inside the adopted envelope. Slosh, ullage motion, reverse-acceleration settling, and feed transients are not resolved. The settled-column approximation is retained through coast and return; restart settling is a separate scenario assumption.

The dry allocation is shell 8000 kg at 21.0 m, fins 1520 kg at 40.3 m, engines 4230 kg at 0.452 m, and folded legs 2000 kg at 6.66 m, using published group estimates with equivalent shapes [8, Table 2]. The remaining 6450 kg closes the 22200 kg budget and is assumed uniformly distributed over 4.0–40.8 m. This yields nominal dry CG 17.521 m. Moving the residual group's centroid anywhere within that interval gives conditional dry-CG bounds 12.18–22.87 m; they are allocation bounds, not statistical confidence limits.

The shell/residual groups use thin-shell equivalents and localized groups use equivalent disks. The liquid columns and loaded Dragon use uniform solid cylinders. Dry-group placements stay fixed; deployed aerodynamic/contact geometry does not separately relocate the dry leg mass. The initial combined vehicle has

CG station 29.3704 m and inertia diagonal approximately $[9.47993 \times 10^5, 1.13857 \times 10^8, 1.13857 \times 10^8]$ kg m^2. These are model outputs (E/A), not measured CRS-10 mass properties.

3.3.2 Component aggregation and analytical derivatives

A `MassComponent` describes one part or liquid about its own centre. `MassProperties` describes the attached vehicle about its current CG. With component mass m_i, axial centroid c_i, local inertia $\mathbf{J}_{i,c}$, and offset $\boldsymbol{d}_i = (c_i - c)\boldsymbol{e}_x$, Equation (5) gives the combined properties:

$$\begin{aligned} m &= \sum_i m_i, \qquad c = \frac{\sum_i m_i c_i}{m}, \\ \mathbf{J}_C &= \sum_i \left[\mathbf{J}_{i,c} + m_i \mathrm{S}(\boldsymbol{d}_i)\right], \qquad \mathrm{S}(\boldsymbol{d}) = (\boldsymbol{d}^\mathsf{T}\boldsymbol{d})\mathbf{I} - \boldsymbol{d}\boldsymbol{d}^\mathsf{T}. \end{aligned} \tag{5}$$

Here $\mathbf{I}$ is the identity matrix. The implementation differentiates both component properties and parallel-axis offsets. Equation (6) states the first derivatives; second CG/mass derivatives are also evaluated analytically for the motion and sensor interfaces:

$$\begin{aligned} \dot{c} &= \frac{\sum_i (\dot{m}_i c_i + m_i \dot{c}_i) - \dot{m} c}{m}, \\ \dot{\mathbf{J}}_C &= \sum_i \left[\dot{\mathbf{J}}_{i,c} + \dot{m}_i \mathrm{S}(\boldsymbol{d}_i) + m_i \dot{\mathrm{S}}(\boldsymbol{d}_i)\right]. \end{aligned} \tag{6}$$

Axial placement makes the parallel-axis addition $\mathrm{diag}(0, d_i^2, d_i^2)$. Thus Body I_{xx} is axial and I_{yy}, I_{zz} are transverse. Figure 3 shows this same operation in code, and Figure 4 shows its first-stage-only inventory dependence. The late change in CG trend follows the changing weights of the two liquid centroids and the fixed dry allocation.

mass_properties.py | lines 157-169

```
157     inertia = np.zeros((3, 3))
158     inertia_rate = np.zeros((3, 3))
159     for c in components:
160         distance = c.center_axial_m - center
161         distance_rate = c.center_rate_mps - center_rate
162         parallel = np.diag([0.0, distance**2, distance**2])
163         parallel_rate = np.diag([0.0] + [2 * distance * distance_rate] * 2)
164         inertia += c.inertia_at_center_kgm2 + c.mass_kg * parallel
165         inertia_rate += (
166             c.inertia_rate_at_center_kgm2ps
167             + c.mass_rate_kgps * parallel
168             + c.mass_kg * parallel_rate
169         )
```

Figure 3 Exact parallel-axis and derivative accumulation in `mass_properties.py`. Each local inertia is shifted to the combined CG once. Source line numbers and file hash are retained with the figure provenance.

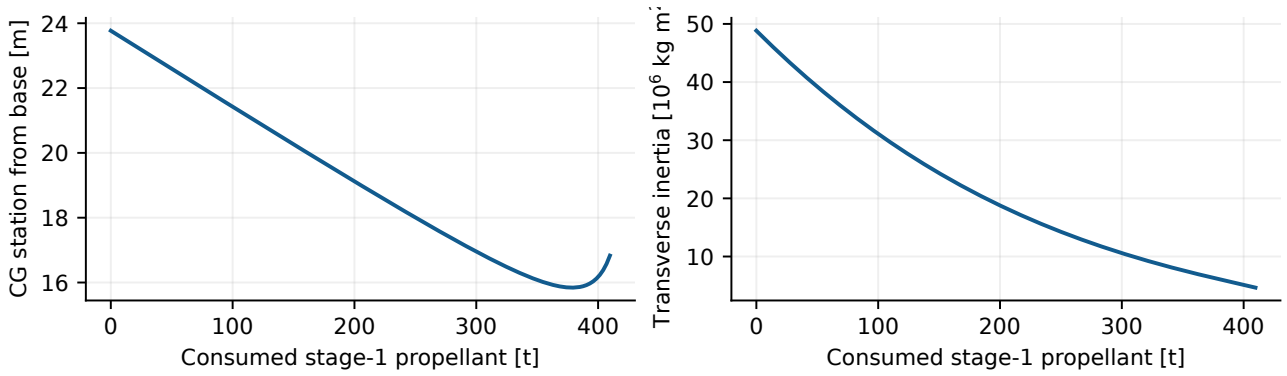


Figure 4 Stage-1-only mass-property sweep from full loading to the fixed-ratio limiting inventory. The curves evaluate the adopted component model against consumed propellant; they are neither measured flight properties nor a separate trajectory simulation.

3.4 Equations of motion and load closure

The changing-mass model uses fixed nozzle-exit stations and balanced transverse outflow. For outlet j, $Q_j \geq 0$ is outward flow, $\mu_j = -Q_j$ its signed mass-removal rate, $\dot{\mu}_j = -\dot{Q}_j$, and $\boldsymbol{d}_j = \boldsymbol{s}_j - \boldsymbol{c}$ the exit arm. $\mathbf{G}_j$ is the outlet's second-moment-per-unit-mass tensor in m^2. The open-system balance retains explicit flow terms, consistent with control-volume rocket dynamics [18].

Equations (7) and (8) are the implemented CG balances. $\boldsymbol{a}_C^B$ is the external-frame coordinate acceleration expressed in Body components; $\boldsymbol{g}^B$ and $\boldsymbol{a}_{\mathrm{rot}}^B$ are supplied separately by the environment:

$$m\boldsymbol{a}_C^B = \boldsymbol{F}^B + m\boldsymbol{g}^B + m\boldsymbol{a}_{\mathrm{rot}}^B + \sum_j \left[2\mu_j(\boldsymbol{\omega}^B \times \boldsymbol{d}_j^B - \dot{\boldsymbol{c}}^B) + \dot{\mu}_j \boldsymbol{d}_j^B\right], \tag{7}$$

$$\mathbf{J}_C\dot{\boldsymbol{\omega}}^B = \boldsymbol{M}_C^B - \boldsymbol{\omega}^B \times (\mathbf{J}_C\boldsymbol{\omega}^B) - \dot{\mathbf{J}}_C\boldsymbol{\omega}^B + \sum_j \mu_j[\mathrm{S}(\boldsymbol{d}_j^B) + \mathbf{G}_j]\boldsymbol{\omega}^B. \tag{8}$$

Pressure-corrected thrust already contains the exhaust reaction; another Qv_e force is not added. The remaining flow terms describe reference migration and angular-momentum outflow. The current closure requires axial CG migration, fixed exits, and balanced transverse exhaust centroids; internal swirl and moving nozzle boundaries are omitted. The code verifies consistency between consumption and mass derivatives before evaluating these balances.

Because the state is stored in current Body axes, Equation (9) includes basis-transport terms. The quaternion uses the same approximate relative-rate convention as Section 2:

$$\begin{aligned} \dot{\boldsymbol{r}}_C^B &= \boldsymbol{v}_C^B - \boldsymbol{\omega}^B \times \boldsymbol{r}_C^B, & \dot{\boldsymbol{v}}_C^B &= \boldsymbol{a}_C^B - \boldsymbol{\omega}^B \times \boldsymbol{v}_C^B, \\ \dot{\boldsymbol{q}}_{BE} &= \tfrac{1}{2}\boldsymbol{q}_{BE} \otimes [0, (\boldsymbol{\omega}^B)^{\mathsf{T}}]^{\mathsf{T}}, & \dot{m}_{p,k} &= -Q_k. \end{aligned} \tag{9}$$

The integrator temporarily uses external translational components within RK4, then returns the accepted state in Body components. This avoids artificial radial interpolation error from rotating an Earth-scale stored vector.

Structural-base and CG states are related by Equation (10), with base O and current CG station $\boldsymbol{c}^B$:

$$\boldsymbol{r}_C^B = \boldsymbol{r}_O^B + \boldsymbol{c}^B, \qquad \boldsymbol{v}_C^B = \boldsymbol{v}_O^B + \boldsymbol{\omega}^B \times \boldsymbol{c}^B + \dot{\boldsymbol{c}}^B. \tag{10}$$

The same transformation applies to a fixed sensor station after replacing the base-to-CG displacement appropriately. `to_state` and `apply_state` use matching flow/rate inputs. Their initial stationary-base condition can therefore have a nonzero geometric-CG velocity from depletion, without implying motion of the sensor mount.

3.4.1 Moment inputs and actuator representation

The moment $\boldsymbol{M}_C^B$ includes aerodynamic moments and an A-class ideal control couple. The active `propulsion.py` relation keeps thrust along Body $+x$ and rejects nonzero gimbal commands; the stored gimbal limit is not an implemented TVC actuator. For `ascent.py`, Equation (11) maps desired Body angular acceleration $\boldsymbol{\alpha}_{\mathrm{cmd}}$ to its added control moment:

$$\boldsymbol{M}_{\mathrm{ideal}} = \mathbf{J}_C\boldsymbol{\alpha}_{\mathrm{cmd}}, \qquad \mathbf{J}_C^{-1}\boldsymbol{M}_{\mathrm{ideal}} = \boldsymbol{\alpha}_{\mathrm{cmd}}. \tag{11}$$

The cancellation in this control contribution is deliberate model inversion, not disappearance of physical inertia or evidence that aerodynamic stability removes the need for TVC. Inertia still determines the requested torque and the response to other loads. The ascent couple is unlimited. `recovery.py` adds gyroscopic feedforward and aerodynamic-moment compensation before fin allocation and moment saturation; changing-inertia and exhaust-flow terms remain in the dynamics.

Thus the scenario demonstrates vehicle propagation with declared moment authority, rather than validated nozzle deflection, gimbal dynamics, or actuator stability margins. A physical TVC model would instead link commanded deflection, actuator limits and dynamics, thrust direction, and the CG moment arm. Flight-command and allocation details belong in the mission-execution chapter; this distinction defines what the moment input represents.

3.5 Initial state and flight-phase definition

The structural base starts at the launch location of Section 2, with zero velocity relative to rotating Earth and zero Body-relative angular rate. The launch quaternion sets Body $+x$ upward; inventories are the full values in Table 3. Commands, configuration transitions, and continuous integration then generate all subsequent states.

Table 6 defines the mission sequence without substituting simulated event times for input conditions. The official schedule labels its entries approximate. Adopting a published time as a simulation command is an A-class choice even when the published schedule itself is authoritative [1].

Transition	Definition in the selected scenario
Liftoff / ascent	Initial attached vehicle; prescribed ascent commands.
Shutdown begins	140 s; project one-second smooth ramp.
MECO / stack coast	141 s; thrust and flow reach zero.
Stage separation	144 s; attached parent, engines off, zero flow and flow derivative.
Upper-stage ignition	152 s; independent upper-child commands.
Boostback ignition	161 s; detached booster and fuel/attitude/readiness guards.
Boostback end / return coast	Miss, fuel, or duration condition followed by smooth shutdown.
Entry ignition	Descending through 45 km with restart guards.
Entry end / aerodynamic descent	$500\,\mathrm{m\,s^{-1}}$ speed target or planning cap, then shutdown.
Landing ignition	Project stopping-clearance prediction with restart guards.
First contact	Lowest assumed deployed leg tip reaches the ground sphere; evaluate landing criteria.
Upper-stage cutoff	545 s; prescribed upper-child shutdown.

Table 6 Physical phase definitions and command references. Entry/landing conditions and return planning criteria are project assumptions. Their actual execution times and numerical results belong to the simulation-results analysis.

First contact uses four assumed leg tips 1.0 m below the base at 3.5 m radius. A successful contact additionally requires deployed legs, vertical speed at most $2\,\mathrm{m\,s^{-1}}$, horizontal speed at most $1\,\mathrm{m\,s^{-1}}$, tilt at most 5°, angular-rate magnitude at most $1\,\mathrm{deg\,s^{-1}}$, and target error at most 10 m. These are A-class project limits, not certified Falcon 9 requirements. The target is an assumed point 14 km south and 6 km east of launch on the ground sphere, rather than surveyed LZ-1 coordinates.

For normalized ramp time $\xi = (t - t_s)/\Delta t$, Equation (12) supplies a smooth shutdown amplitude and its time derivative on $0 \le \xi \le 1$:

$$\gamma = 1 - 10\xi^3 + 15\xi^4 - 6\xi^5, \qquad \dot{\gamma} = -\frac{30\xi^2(1-\xi)^2}{\Delta t}, \qquad \Delta t = 1\,\mathrm{s}. \tag{12}$$

The envelope is constant outside the ramp, with zero endpoint slope and curvature. Scaling thrust and flow together while supplying the analytical flow derivative avoids an unmodeled discontinuous-flow impulse. The duration is a numerical assumption, not a measured engine transient. Return planning budgets, steering laws, and actuator allocation are kept in mission execution rather than embedded in the vehicle equations.

3.6 Separation and inherited child states

The separation map assumes zero impulse and no discarded mass. Both children inherit the parent's attitude and angular rate. For a cut offset $\boldsymbol{\ell}^B$ from the parent structural base, the lower base retains the parent's base state and the upper base follows Equation (13):

$$\boldsymbol{r}^B_{O,u} = \boldsymbol{r}^B_{O,p} + \boldsymbol{\ell}^B, \qquad \boldsymbol{v}^B_{O,u} = \boldsymbol{v}^B_{O,p} + \boldsymbol{\omega}^B \times \boldsymbol{\ell}^B. \tag{13}$$

Upper-child component stations are rebased consistently, while Dragon stays on that child. Different child CG coordinates are expected because the tracked points have changed. Position/velocity continuity is checked at fixed structural points and at the recombined mass centre.

The conservation audit uses angular momentum about the parent CG. With child CG offset $\boldsymbol{\rho}_k$ and velocity difference $\Delta\boldsymbol{v}_k$ from the parent CG, Equation (14) includes spin and displaced-CG motion:

$$\boldsymbol{H}_{\text{after}} = \sum_k \left[\mathrm{J}_k \boldsymbol{\omega}_k + \boldsymbol{\rho}_k \times (m_k \Delta\boldsymbol{v}_k) \right]. \tag{14}$$

The map compares this value with the parent spin angular momentum, and also checks mass, linear momentum, kinetic energy, recombined CG, and structural-point continuity. Relative conservation tolerance is 10^{-10}; position and velocity continuity tolerances are 10^{-8} m and $10^{-8}\,\mathrm{m\,s^{-1}}$; attitude/rate checks use 10^{-12} in their respective norms. These are numerical consistency criteria for the assumed map, not physical separation accuracy.

Afterward, stage 1 excludes stage 2 and Dragon from its mass properties, and the upper child excludes stage 1. Both use inherited states and separate subsequent commands on the same clock. The upper stage retains the uniform propellant/Dragon approximations and its separate drag provider, rather than automatically selecting the declared `second_stack` aerodynamic configuration. Dragon remains attached over the 600 s demonstration interval. The sequence tests continuous configuration/state handling; public validation of the complete return and upper-stage command laws is outside the available ascent reference.

4 Environment Models

The environment module (`Environement.py`) is a pure service layer: it supplies the world the vehicle flies through and never calls back into vehicle, dynamics or control code. Its services are the standard atmosphere, point-mass gravity with its gradient, the acceleration of the rotating Earth-fixed frame, the wind, and the ground reference; all evaluations are stateless functions of the current state. Following the project convention, any state outside a modeled domain is rejected with an error, never extrapolated.

4.1 Atmosphere

The atmosphere follows the US Standard Atmosphere 1976 [10]: seven gradient layers between sea level and the mesopause at 84.852 km, reproduced layer by layer. Within a layer with base temperature T_b, base pressure p_b and lapse rate L, and in the isothermal case, the pressure is

$$p(h) = p_b \left(\frac{T_b + L\,(h - h_b)}{T_b} \right)^{-\frac{g_0}{R L}} \quad (L \neq 0), \qquad p(h) = p_b \, \exp\left(-\frac{g_0\,(h - h_b)}{R\,T_b} \right) \quad (L = 0), \tag{15}$$

with $g_0 = 9.80665\,\mathrm{m/s^2}$ and the gas constant of dry air $R = 287.05287\,\mathrm{J/(kg\,K)}$. Density and speed of sound follow the ideal-gas and isentropic relations, $\rho = p/(R\,T)$ and $a = \sqrt{\gamma R T}$ with $\gamma = 1.4$. Sea-level values are $T = 288.15\,\mathrm{K}$, $p = 101\,325\,\mathrm{Pa}$, $\rho = 1.225\,\mathrm{kg/m^3}$.

Above the mesopause the model deliberately stops claiming physics: an isothermal tail at $T_m = 186.946\,\mathrm{K}$ decays exponentially with a scale height of $R\,T_m/g_0 \approx 5.47\,\mathrm{km}$ toward numerical vacuum (density below $10^{-150}\,\mathrm{kg/m^3}$ at the 2 000 km envelope ceiling; $p(200\,\mathrm{km}) \approx 2.7 \times 10^{-10}\,\mathrm{Pa}$). The envelope 0–2 000 km is the mission scope; altitudes outside raise a domain error instead of a number. Figure 5 shows the model against the published USSA 1976 checkpoint values that the test suite pins (5–80 km, temperatures, pressures and densities at 5×10^{-5} relative tolerance); the environment suite of 41 tests additionally checks layer-boundary continuity, monotone decay, exact tail scale height, hydrostatic balance and the domain rejection behavior.

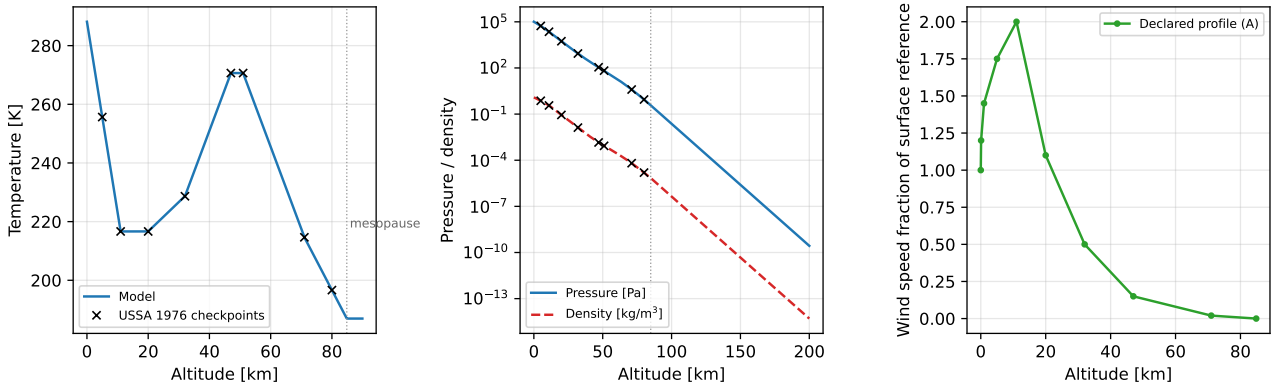


Figure 5 Environment model. Left: layer temperatures to the mesopause with published USSA 1976 checkpoint values. Center: pressure and density (logarithmic) to 200 km, showing the isothermal tail above the mesopause; black markers are the published checkpoints. Right: the declared wind-profile scale factor of the surface reference wind (class A), which is zero above the modeled atmosphere; the nominal mission runs in calm air.

4.2 Gravity and rotating-frame accelerations

Gravity is the reduced Newtonian point-mass model. With the Earth radius $R_E = 6\,371\,\mathrm{km}$ and the configured surface gravity g_0, the gravitational parameter is $\mu = g_0 R_E^2$, so the gravitational acceleration at the launch site is the configured value by construction:

$$\mathbf{a}_g = -\frac{\mu}{\|\mathbf{r}\|^3}\,\mathbf{r}. \tag{16}$$

Gravity never appears as a force; Equation (16) is added once as an acceleration next to thrust over mass. The module also provides the 3×3 gravity-gradient map $\nabla_{\mathbf{r}}\mathbf{a}_g = (\mu/\|\mathbf{r}\|^3)\,(3\,\hat{\mathbf{r}}\hat{\mathbf{r}}^{\mathsf{T}} - \mathbf{I})$, used only in the navigation filter's prediction Jacobian (Section 7), not for torques. Oblateness (J_2) is out of scope.

The external frame is Earth-centred and launch-aligned (the shared conventions are fixed in the architecture chapter of this report); it does not rotate for the flight mechanics themselves. Earth rotation enters translation only, through the separate frame acceleration

$$\mathbf{a}_{\mathrm{fr}} = -2\,\boldsymbol{\omega}_E \times \mathbf{v} \;-\; \boldsymbol{\omega}_E \times (\boldsymbol{\omega}_E \times \mathbf{r}), \tag{17}$$

the Coriolis and centrifugal terms for velocity relative to the rotating Earth, with the rotation vector $\boldsymbol{\omega}_E = \Omega\,(\cos\varphi,\ 0,\ \sin\varphi)$ at launch latitude φ. The nominal mission uses the sidereal rate $\Omega = 7.2921150 \times 10^{-5}$ rad/s at $\varphi = 28.6°$; setting $\Omega = 0$ disables the terms. The frame accelerations, and their analytical gradients, are supplied once per integration step next to gravity. The angular equations and the gyro model retain the project's approximate Earth-rate treatment: no complete inertial-rate coupling is claimed.

4.3 Wind and ground reference

The wind model is a surface reference vector (ECEF components in the mission configuration) that can be rotated about the launch-site vertical, scaled by a declared altitude profile: the scale factor of the surface reference grows through the boundary layer, peaks at 2.0 around the 11 km jet-stream core, decays through the stratosphere and reaches exactly zero at the mesopause — there is no wind above the modeled atmosphere (Figure 5, right). The profile is a declared stylization (class A). There are no gusts, no time dependence and no direction shear, and the wind enters the physics only through the relative-velocity term of the aerodynamics model. The nominal full-mission run uses calm air: the mission configuration sets no surface reference wind, so the default zero vector applies throughout.

The ground is a bare sphere of radius R_E; altitude is measured radially above it, with an optional reference-level shift, and supports a station offset from the center of gravity (used, for example, for engine-exit altitude). Terrain and contact forces are out of scope; contact is evaluated at the mission level through the leg-tip geometry discussed with the landing criteria in the mission-execution chapter of this report.

Assumptions and limitations The environment is dry air only, with a strictly layered USSA profile; gravity is spherical with no oblateness; the wind is a declared stylization without gusts, time dependence or direction shear; the ground is a bare sphere without terrain; and the frame-acceleration treatment is exact for translation while the attitude chain carries only the approximate Earth-rate treatment noted above. These scope choices (class A) apply uniformly to ascent and return; no environment parameter was fitted to flight data. The module validates its configuration at construction — non-finite gravity, radius, ground offset or wind, non-positive radius, a negative rotation rate or a latitude beyond ±90° all raise an error before flight, not mid-integration.

5 Aerodynamics

The aerodynamic model (`Aerodynamics.py`, data in `data/aero_configurations.json`) supplies aerodynamic forces and moments in current Body components, Body $+x$ toward the nose, stations measured from the structural datum of the active configuration. One model instance serves the whole flight through five declared vehicle configurations, switched by the mission layer at phase transitions; every coefficient entry in the data file carries an evidence and uncertainty record (the evidence-class convention of Section 3).

5.1 Flow state and force build-up

All aerodynamic terms act on the relative flow velocity

$$\mathbf{v}_{\text{rel}} = \mathbf{v}_{\text{CG}} - \dot{\mathbf{c}} - \mathbf{w}, \qquad \alpha = \operatorname{atan2}\big(\lfloor \mathbf{v}_{\text{rel},yz} \rfloor,\ v_{\text{rel},x}\big), \tag{18}$$

where $\dot{\mathbf{c}}$ is the axial center-of-gravity migration rate (Section 3, stage depletion) and $\mathbf{w}$ the wind in Body components (Section 4). The total flow angle α spans 0 (nose-first) to π (engine-first); every direction in the lateral plane is handled, not only a pitch plane. The dynamic pressure is $\bar{q} = \frac{1}{2}\,\rho\, v_{\text{rel}}^2$ with the atmosphere state of Section 4.

The model decomposes the load into an axial, a potential-normal and per-segment crossflow terms:

$$F_x = -\,\bar{q}\, S_{\text{ref}}\, CA_0(M)\ \cos\alpha\,|\cos\alpha|, \qquad N = \bar{q}\, S_{\text{ref}}\, C_{N\alpha}\ \sin\alpha\,|\cos\alpha|, \tag{19}$$

$$D_i = \bar{q}\ C_{Dc,i}\, S_i\ \sin^2\alpha \qquad \text{per side segment } i, \tag{20}$$

with reference area and length per configuration (Table 7). The normal force acts along the negative lateral relative-velocity direction at the configuration's windward-end station (57.6 m full stack, 41.0 m booster, 17.3 m second stack), mirrored about the center-of-gravity station for retro flow; crossflow segments act at their side-area centroids. For `booster_landing` the body segment (width 3.66 m, $C_{Dc} = 1.1$) is supplemented by the combined grid-fin segment (38.6–40.6 m, 2.4 m, $C_{Dc} = 1.0$) and the deployed-leg segment (0–3.6 m, 1.6 m, $C_{Dc} = 1.2$).

The $\cos\alpha\,|\cos\alpha|$ factor appears in Equation (19) on purpose: the axial force scales with the axial dynamic-pressure share and reverses direction for retro flight, and the grid-fin effectiveness below scales the same way, so a fixed deflection command reverses its steering effect in engine-first flight and vanishes at $\alpha = 90°$. Station moments are computed once as $(\text{station} - \text{CG}) \times \mathbf{F}$ and returned as a force through the center of gravity plus a free moment, so the load interface cannot double-count. Windward-end pressure centers toward the nose are destabilizing in nose-first flight; deployed hardware near the trailing end stabilizes tail-first descent. The model reproduces this textbook behavior from the stated geometry — it does not stabilize the vehicle on its own.

Rotary damping closes the moment set:

$$\mathbf{M}_{\text{damp}} = c\ \frac{\bar{q}\, S_{\text{ref}}\, L_{\text{ref}}^2}{2\, v_{\text{rel}}}\ \boldsymbol{\omega}, \tag{21}$$

with pitch and yaw tied by symmetry and roll treated separately (declared coefficients c: -10 full stack, -20 booster, roll -3 with fins deployed; a declared zero means roll damping is modeled as unmodeled for that configuration).

5.2 Configurations and axial curves

Figure 6 shows the adopted axial coefficient curves $CA_0(M)$. Only the full-stack curve is sourced: a raster readout of Pezent's RasAeroII analysis (0° angle of attack, full Falcon-9-like stack with fairing) [7], whose transonic peak $CA_0 \approx 1.16$ near $M \approx 1.06$ carries the worst readout uncertainty. Beyond Mach 7 the curve continues as a declared slow hypersonic decay to $CA_0 = 0.75$ at Mach 10, where the vehicle flies above roughly 55 km and axial drag stays below about 0.5 % of thrust. All other curves are declared engineering estimates: a blunt-ended cylinder for the booster ($CA_0 = 0.42$ subsonic, raised to 0.67 by the flat fin/leg increment), a sleeker estimate for stage 2 plus Dragon (0.32 subsonic), uncertain to roughly ±50 %. They close the simulation loop and produce plausible dynamics; they are not good enough to predict quantitative trajectories beyond the sourced regime. Replacing them is a data change in the JSON, not a code change.

Configuration	Phase	S_{ref} [m^2]	L_{ref} [m]	Axial curve / fins
`full_stack`	launch to separation	21.08	62.4	sourced M 0–7; declared M 7–10
`booster_alone`	coast, boostback	10.52	42.6	declared (±50 %)
`booster_return`	entry burn, descent	10.52	42.6	declared; four fins
`booster_landing`	landing burn	10.52	42.6	declared; four fins and legs
`second_stack`	stage-2 body (declared)	10.52	19.8	declared, sleeker estimate

Table 7 The five aerodynamic configurations. Station datum is the configuration's own structural base. Fins deploy after boostback; the legs deploy at landing ignition, which is represented by the segment difference between `booster_return` and `booster_landing`.

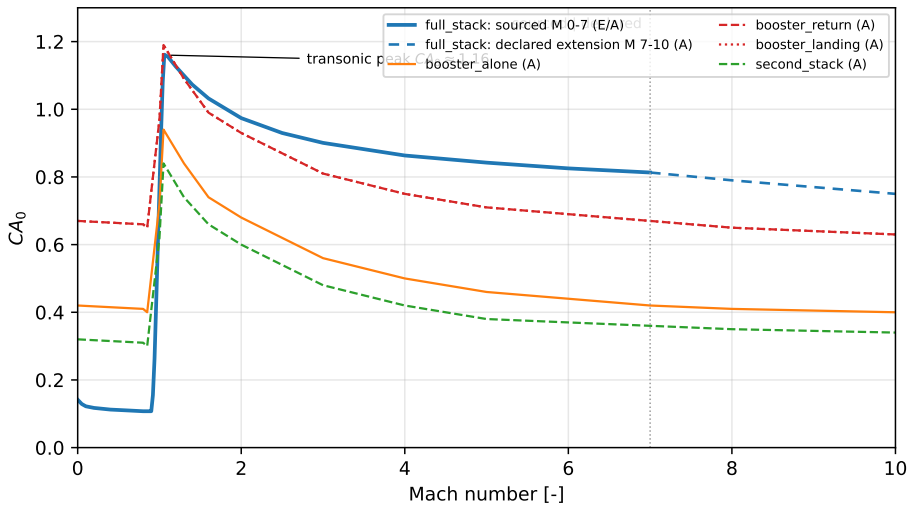


Figure 6 Axial drag coefficient $CA_0(M)$ of the five configurations. Only the full-stack curve to Mach 7 is sourced; everything else is a declared estimate (evidence classes in the legend). The transonic peak of the full stack reaches $CA_0 \approx 1.16$.

5.3 Grid fin control

The `booster_return` and `booster_landing` configurations declare four grid fins at intercardinal azimuths (45°, 135°, 225°, 315° about Body $+x$), mounted at station 39.6 m on a 1.83 m pivot radius; each fin is modeled as a porous lattice panel of 1.65 m^2 with lift slope $c_{\ell,\delta} = 1.2/\mathrm{rad}$ (about half of a solid panel of aspect ratio near one, because of lattice porosity) and a travel limit of ±0.35 rad (≈ ±20°). A positive deflection of fin i produces the force

$$\mathbf{F}_i = \bar{q}\, S_{\mathrm{fin}}\, c_{\ell,\delta}\ \delta_i \cos\alpha\, |\cos\alpha|\ \big(0, -\sin\varphi_i, \cos\varphi_i\big), \tag{22}$$

applied at the fin's three-dimensional mount station, so the axial lever carries pitch/yaw and the radial offset carries roll — the moment combinatorics are automatic: equal deflections give pure roll, opposite differential pairs give pure pitch, a single fin gives both. At the declared values, full deflection at Mach 3 and 15 km commands about 1.4 MN m of pitch and 385 kN m of roll, an order of magnitude less above 30 km — consistent with grid fins mattering mainly below entry-burn altitudes. Command shaping (rates, travel dynamics, deadbands) belongs to the actuator layer of the control chain and is not part of this model: the aerodynamics accept instantaneous validated commands only. Mach- and angle-of-attack-dependent fin effectiveness, interference with the passive crossflow segments, lattice state and the deflection axial-drag increment are declared unmodeled.

5.4 Domain and verification

Following the project convention, the model rejects rather than extrapolates: Mach outside the adopted curve (all configurations end at $M = 10$), unknown configuration names, command shape or travel violations, and construction-time station violations all raise errors. Within the domain, declared-estimate coefficients do not fail — they produce plausible numbers with real uncertainty.

The test suite (`tests/test_aerodynamics.py`, 42 tests) pins the conventions: the zero-angle drag baseline, normal-force direction and the windward-end mirror convention in both flow directions, the damping formula term by term, the physical invariants (energy extraction at every angle, dissipative damping, wind entering only through the relative velocity, continuity at $\alpha = 90°$), the fin combinatorics of Equation (22) at all azimuths including retro reversal, and the loading of all five configurations from the JSON with the sourced knot intact, plus a two-second vertical propagation through the shared dynamics core.

6 Sensor Models and Measurement Generation

Four sensor front-ends generate measurements from the truth trajectory: an inertial measurement unit (IMU), a global navigation satellite system (GNSS) receiver, a light detection and ranging (LiDAR) altimeter, and a star tracker (`sensors.py`). Sampling occurs only at accepted integration times whose 0.01 s step is an integer multiple of every sensor period, so all rates in Table 8 are hit exactly; no random numbers are drawn during integration trial evaluations, and a fixed seed reproduces a run sample for sample. The navigation-consumer side of these streams is the filter of Section 7; this section covers measurement generation only. All specs are placeholder research assumptions (classes A/C), not certified Falcon 9 hardware.

Sensor	Rate	Mounting	Error model
IMU	100 Hz	station $[38, 0, 0]$	accel $\sigma = 0.01\,\mathrm{m/s^2}$, bias $[0.05, -0.03, 0.02]\,\mathrm{m/s^2}$, scale factor 10^{-3}, bias random walk 10^{-4}; gyro $\sigma = 10^{-4}\,\mathrm{rad/s}$, bias $[0, 0, 10^{-4}]\,\mathrm{rad/s}$; misalignment and temperature models supported
GNSS	10 Hz	antenna $[41, 0, 0]$	pos $\sigma = [1.5, 1.5, 2.5]$ m plus Gauss-Markov $\sigma = [0.6, 0.6, 1.0]$ m at $\tau = 300$ s; vel $\sigma = 0.1$ m/s; latency 0.1 s; outages and dropouts supported
LiDAR	10 Hz	station $[0.5, 0, 0]$, boresight $-x$	slant range 1–1 800 m, $\sigma = 0.05\,\mathrm{m} + 10^{-4}\,R$
Star tracker	4 Hz	boresight 60° from $+x$ toward $+y$	cross-boresight $\sigma = 5$ arcsec, boresight roll $\sigma = 40$ arcsec; latency 0.1 s

Table 8 Sensor front-ends and adopted settings; the stations are Body coordinates. Every entry is a placeholder research assumption (class A/C), not a certified Falcon 9 sensor.

6.1 Inertial measurement unit

The IMU measures specific force and angular rate at its fixed structural station. The ideal specific force includes the station kinematics of the dynamics core — the angular-rate and center-of-gravity-migration terms of moving the measurement point off the center of gravity — so the accelerometer senses what a real strapped-down unit at that station would sense. Per sample, the measured pair is

$$\mathbf{f}_{\mathrm{meas}} = \mathbf{A}\,\mathbf{f}_{\mathrm{true}} + \mathbf{b}_a + \mathbf{b}_{a,T} + \mathbf{b}_{a,\mathrm{rw}} + \mathbf{n}_a, \qquad \boldsymbol{\omega}_{\mathrm{meas}} = \mathbf{G}\,\boldsymbol{\omega}_{\mathrm{true}} + \mathbf{b}_g + \mathbf{n}_g, \tag{23}$$

with per-axis scale-factor/misalignment matrices $\mathbf{A}, \mathbf{G} = \mathbf{I} + \mathrm{diag}(\cdot) + \mathbf{M}$ (zero diagonal of $\mathbf{M}$), constant biases, temperature coefficients about a 20 °C reference, bias random walks driven per sample as $\mathcal{N}(0, \sigma_{\mathrm{rw}}^2\,\Delta t)$, and white noise at the sigmas of Table 8. The nominal mission uses the biases, scale factor and random walk listed in the table, with misalignment, temperature and gyro scale-factor terms at zero (modeled but switched off). The gyro measures the Body rate relative to the rotating Earth-centred Earth-fixed frame (ECEF) frame, consistent with the quaternion transport equation the navigation filter propagates.

6.2 GNSS, LiDAR and star tracker

The GNSS receiver provides antenna position and velocity in the external frame, transformed from the truth state with the full lever-arm kinematics (antenna station minus IMU station, angular-rate cross term and center-of-gravity migration). A 0.1 s latency model feeds a truth-side history buffer that is linearly interpolated at the delayed epoch, so the emitted sample is a measurement of the past state, not of the present one. The position error is a first-order Gauss-Markov process beside the white noise (updating per sample with the configured 300 s correlation time), modelling the slow, correlated component absent from naive white-noise GNSS error. First-fix delays, scheduled outages and random dropouts are supported and reported through a validity flag; the nominal run has none of them, and every sample is valid.

The LiDAR measures slant range to the spherical ground along its $-x$ boresight. Validity is geometric: the analytic line–sphere intersection must exist, point downward and land inside the declared 1–1 800 m envelope; invalid geometry is reported as an invalid sample, never extrapolated. Range noise is $0.05\,\mathrm{m} + 10^{-4}R$.

The star tracker provides absolute attitude as a sensor-frame-to-Earth-centred inertial frame (ECI) quaternion. The ECI frame is defined to coincide with the launch-aligned ECEF frame at $t = 0$, and the Earth rotation carries the conversion, so the injection of rotation into the attitude chain is explicit rather than implicit. Noise is 5 arcsec cross-boresight and 40 arcsec about the boresight per exposure, with a 0.1 s latency applied on the attitude history. Availability is physical: the unit is blinded above $2°/\mathrm{s}$ body rate, below 60 km altitude (dense atmosphere), inside the $20°$ Earth-limb exclusion, and optionally inside a $40°$ solar exclusion; first-fix, outage and dropout faults are supported (nominally off). The reported sample always carries its blinding reason, making the availability record auditable in the run output.

Assumptions and limitations All noise, bias and latency values are declared placeholders, not datasheet values; the GNSS model has no ionospheric, tropospheric or multipath terms; the LiDAR assumes the bare sphere of Section 4 as ground; the star tracker mounting is a placeholder that keeps the boresight sky-pointing for an upright vehicle, and vehicle-generated obscuration (plumes, stage interfaces) is not modeled.

7 Navigation Architecture and State Estimation

The navigation filter, `kalman.py`, is a multiplicative error-state multiplicative extended Kalman filter (MEKF): inertial prediction on the IMU stream, corrected by the GNSS, LiDAR and star-tracker measurements of Section 6. The estimated state is referenced to the IMU station — its position and velocity in the external frame — so the lever arms of the other sensors are constant Body vectors and no state re-referencing occurs when the center of gravity migrates during burn. The nominal 24-state configuration is $\mathbf{x} = [\mathbf{p}, \mathbf{v}, \mathbf{b}_a, \mathbf{q}, \mathbf{b}_g, \mathbf{s}_f, \mathbf{m}]$: IMU-station position and velocity, accelerometer bias, the attitude quaternion from Body to ECEF (Hamilton, scalar-first), gyro bias, per-axis gyro scale factor and the six off-diagonal gyro misalignment terms (Table 9).

Block	Content	Initial 1σ
$\mathbf{p}$, $\mathbf{v}$	IMU-station position, velocity [m, m/s]	2 m, 0.05 m/s
$\mathbf{b}_a$	accelerometer bias, Body [m/s^2]	0.1
$\mathbf{q}$	attitude (multiplicative error $\delta\boldsymbol{\theta}$, Body [rad])	0.01 rad
$\mathbf{b}_g$	gyro bias, Body [rad/s]	10^{-3}
$\mathbf{s}_f$, $\mathbf{m}$	gyro scale factor, misalignment	10^{-3}

Table 9 Estimated-state blocks and initial standard deviations. The initial position error is a seeded $[1, -1, 0.5]$ m; the attitude starts at truth with a 0.01 rad prior unless declared otherwise.

7.1 Prediction

Each 100 Hz IMU sample advances the strapdown state. The gyro measurement is corrected with the current calibration estimate, $\hat{\boldsymbol{\omega}} = \hat{\mathbf{G}}^{-1}(\boldsymbol{\omega}_{\text{meas}} - \hat{\mathbf{b}}_g)$, and the attitude is propagated by its quaternion transport equation with the trapezoidal mean rate over the interval; the mid-interval attitude rotates the bias-corrected specific force into the external frame alongside the modelled gravity and rotating-frame accelerations of Section 4, whose analytical gradients linearize that model in the filter. The error dynamics couple the multiplicative attitude error into the velocity error through $-\mathbf{R}\,[\hat{\mathbf{f}}]_\times \delta\boldsymbol{\theta}$, and the gyro chain drives the attitude error through the bias, scale-factor and misalignment blocks — the calibration is observable only through attitude aiding and sufficient angular motion. Process noise is built from the sensor power spectral densitys (PSDs) ($\sigma_a^2/f_{\text{IMU}}$, $\sigma_g^2/f_{\text{IMU}}$) and the declared bias random walks.

7.2 Measurement updates

All corrections use the Joseph-form covariance update and record a normalized innovation square for monitoring. The GNSS update corrects position and velocity: its 0.1 s latency is compensated by extrapolating the measurement to the current epoch with the estimated acceleration, its lever arm (antenna minus IMU station) is removed with the angular rate, and the measurement Jacobian carries the full attitude, bias and calibration sensitivity of that arm. The LiDAR update is scalar and nonlinear: the measurement is the analytic line–sphere slant range of Section 6, differentiated analytically at the estimated state, with attitude sensitivity through both the arm and the boresight direction; invalid geometry skips the update. The star-tracker update is an absolute attitude correction: the measurement quaternion is converted from ECI to ECEF by the Earth rotation at the exposure epoch, propagated over its latency with the last estimated body rate, and differenced multiplicatively from the estimated attitude; its noise covariance is rotated into the Body frame by the known mounting, and the update is gated by the three-degree-of-freedom normalized innovation squared (NIS) at the 99.9 % level (16.27), with rejects counted rather than silently absorbed.

7.3 Interfaces and diagnostics

The filter consumes its own attitude estimate by default (`use_estimated_attitude`): the prediction rotates the specific force with the estimated quaternion, and every GNSS, LiDAR and star-tracker correction also corrects that attitude. A declared diagnostic mode retains the original nine-state filter with the truth attitude supplied from outside; its results are kept distinct from the gyro-driven results in the results chapter. A second, deliberately unaided baseline propagates the attitude from the raw gyro stream alone (`GyroAttitude`):

starting from the same initial attitude, it carries uncompensated gyro noise and bias, and its logged error isolates how much the aiding buys over pure dead reckoning. One filter instance runs across staging without a reset — the estimated state is IMU-station referenced and the sensor stations are stage-1 hardware, so the booster return needs no state handover; the separated upper stage flies an open-loop truth continuation without a navigation instance.

7.4 Verification and nominal-run behavior

The filter is verified in stages: numeric finite-difference checks confirm the analytic Jacobians of the attitude error with respect to gyro bias, scale factor and misalignment; a closed-loop scenario (a vehicle exercised on the rotating Earth with star-tracker aiding) recovers injected bias, scale-factor and misalignment values; and the cross-module and mission-level suites (115 tests, all green) exercise the integrated chain end to end. In the nominal full-mission run the filter operates in its default gyro-driven configuration: the position error holds 0.881 m root mean square (RMS) (0.918 m in the truth-attitude reference configuration), the aided attitude error stays at 0.456° RMS (peak 1.57°, touchdown 0.012°) while the unaided gyro baseline drifts monotonically to 1.89°, and the z-axis gyro bias is recovered to 1.4 %. The GNSS innovations are statistically consistent (mean NIS 7.58 for six degrees of freedom) and LiDAR delivers 496 valid updates. All 381 valid star-tracker exposures are rejected by the NIS gate: the propagated attitude uncertainty (1σ median 0.118°) understates the actual maneuver-induced error by more than the factor the 99.9 % gate tolerates, so the nominal attitude is gyro-propagated with weak GNSS lever-arm aiding only — the gate outcome and its causes are treated as a result in the results chapter of this report.

Assumptions and limitations All noise and bias parameters are the placeholder research assumptions of Table 8; the gyro calibration block estimates a unit whose adopted true misalignment and scale factor are zero, so its output is a zero-check rather than a calibration demo; GNSS latency compensation assumes constant acceleration across the 0.1 s window; attitude prior and initialization are declared values, not surveyed alignment; and the filter has no protection against unmodeled faults beyond the star-tracker NIS gate — every update it accepts is trusted once inside the gate.

8 Mission Execution and Numerical Implementation

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